KARAN NAIR

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PROFILE

Final-year undergraduate student focused on robot learning. Trained and deployed depth-conditioned Reinforcement Learning (RL) policies on quadruped hardware, with hands-on experience in world models, MPC, and Sim-to-Real transfer. Looking to extend this work towards learned manipulation.

EDUCATION

BITS Pilani, Hyderabad Campus Aug 2023 – May 2027 (expected)
B.E., Electronics and Instrumentation | CGPA: 8.59/10.0 *Hyderabad, India*
Minor: Robotics and Automation | GPA: 9.40/10.0

EXPERIENCE

Strider Robotics (IISc Bangalore Stochastic Robotics Lab spinoff) Jun 2026 – Dec 2026
Research Intern, Advisor: Dr. Shishir Kolathaya *Bengaluru, India*
Platform: MULE3 payload-adaptive quadruped robot

- Train **terrain-adaptive RL locomotion policies** for MULE3 in **NVIDIA IsaacLab**, using a **JEPA-style latent world model** to encode depth images and proprioception into the policy's state.
- Deployed **depth-conditioned RL policies** on the MULE3 robot to **climb stairs and slopes**, with the policy learning to anticipate terrain from depth and raise its feet ahead of stair edges, without using manual gait logic.
- Own the IsaacLab-to-MULE3 **Sim-to-Real loop**: diagnose hardware failures (torque spikes on stairs, missed stair detection, excessive foot lift, reward hacking) and resolve them through depth-camera pipeline fixes, reward shaping, domain randomization, and actuator limit tuning.
- Currently working on extending the same JEPA-based RL policy stack to **quadruped parkour skills** (leaping, climbing, crawling and tilting).

PROJECTS

PIP-Loco: Model-Based RL for Blind Quadruped Locomotion *Jan 2026 – Feb 2026*

- Reimplemented the **PIP-Loco framework (ICRA 2025)** from scratch in Genesis and PyTorch, training a Unitree Go2 for robust blind locomotion in 4 hours on an RTX 3050.
- Implemented an **Asymmetric Actor-Critic** architecture with a **No-Latent Model (NLM) Dreamer**, injecting model-based predictive planning into a reactive RL policy to withstand external velocity perturbations of up to 1.0 m/s.
- Developed a **Temporal Convolutional Network (TCN)** for proprioceptive velocity estimation, enabling accurate command-velocity tracking from a history of onboard sensor data and past actions.
- Built a safe RL training pipeline with strict gradient isolation across three decoupled optimizers and quadratic barrier penalties enforcing torque and velocity limits matched to the Go2's actuators.

The Semantic Manipulator *Nov 2025 – Jan 2026*

- Built a local control stack using **model2vec** and **Logistic Regression** to translate **natural language** commands into robot actions with under 5 ms latency, with no cloud API dependency.
- Implemented a deterministic FSM safety layer that validates every output of the learned classifier against physical constraints before execution.
- Developed a **monocular perception and calibration pipeline**: HSV block segmentation, pinhole depth projection, and an ArUco-based homography transform mapping 2D pixel targets to the robotic arm's base frame.

Real-Time Projectile Interception using a 3-DOF Robotic Arm *Jul 2025 – Oct 2025*

- Built a classical estimation-and-control pipeline in Python for a planar 3-DOF arm to intercept ballistic projectiles in simulation: a **least-squares trajectory estimator** over noisy 3D observations feeding a closed-loop **receding-horizon controller** (custom PyBullet environment).
- Designed a **Fixed-Time MPC** that pre-computes the interception time, eliminating optimization failures and cutting computation to **under 15 ms** for real-time tracking.

TECHNICAL SKILLS

Learning and control: Reinforcement Learning (PPO, asymmetric actor-critic), world models (Dreamer, JEPA), Model Predictive Control, Sim-to-Real transfer
Frameworks and simulators: Python, PyTorch, NVIDIA IsaacLab (built on Isaac Sim), Genesis, PyBullet
Robotics and perception: Robot Operating System (ROS2), OpenCV